



**Joint AEP, CPS, Oncor and CenterPoint
Texas 765-kV STEP: Eastern Backbone
Project – ERCOT Independent Review
Scope**

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RPG Meeting
September 25, 2025

Introduction

- AEP Texas (AEP), CPS Energy (CPS), Oncor Electric Delivery and CenterPoint Energy Houston Electric, LLC (CEHE) submitted the Texas 765-kV STEP: Eastern Backbone Project for Regional Planning Group (RPG) review in July 2025
 - This Tier 1 project is estimated to cost \$9.384 billion and will require a Certificate of Convenience and Necessity (CCN) filing
 - Estimated in-service date (ISD) is between 2030 and 2032
 - This project was identified and included in 2024 Regional Transmission Plan (RTP) in Bexar, Dallas, Bell, Kleberg, Rusk, Wharton, other counties in the North Central, South, Central, Southern, Coast, and East Weather Zones
 - This project aligns with the 2024 RTP 765-kV STEP Core Plan

RPG Study Mode

- Comments were received during the comment period, and the project was placed in study mode on August 8, 2025
- Responses were provided by Joint TSPs and ERCOT to address the consolidated comments received
- Additional reply comments were received by LCRA TSC and TNMP
- Additional responses were provided by the Joint TSPs and ERCOT to address the reply comments
- This project will be under ERCOT Independent Review upon completion of the study mode

Project Proposed by AEP, CPS, Oncor and CEHE

- Construct a new Blu Lacy 765/345-kV Substation and Install;
 - Five (5) 765-kV Circuit Breakers in a ring bus configuration expandable to breaker-and-a-half configuration;
 - Four (4) 765-kV, 300 MVA Switch Shunt Reactors (100 MVA/phase);
 - Two (2) Bus Switch Shunts each with a Circuit Breaker; and
 - Two (2) Line Switch Shunts each with a Circuit Breaker
 - Eleven (11) 345-kV Circuit Breakers in a seven-position (7) breaker-and-a-half configuration; and
 - Three (3) 765/345-kV transformers rated to at least 2,403 MVA Normal and 2,772 MVA Emergency;
- Construct a new Watermill 765-kV Switchyard and Install;
 - Twelve (12) Circuit Breakers in a double-bus, double-breaker configuration; and
 - Unspecified reactive device on 765-kV line

Project Proposed by AEP, CPS, Oncor and CEHE (Cont.)

- At Existing Watermill 345-kV Substation:
 - Install six (6) Circuit Breakers in a breaker-and-a-half configuration
- At Watermill 765/345-kV Substation Install:
 - Three (3) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency
- Construct a new Martin Lake 765-kV Switchyard and Install:
 - Eight (8) Circuit Breakers in a double-bus, double-breaker configuration
 - Unspecified reactive device on 765-kV line
- At Existing Martin Lake 345-kV Substation:
 - Install four (4) Circuit Breakers in a double-bus, double-breaker configuration.
- At Martin Lake 765/345-kV Substation Install:
 - Two (2) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency.

Project Proposed by AEP, CPS, Oncor and CEHE (Cont.)

- Construct a new Hillje 765-kV Substation and Install:
 - Install eight (8) Circuit Breakers in a breaker-and-a-half configuration
- At Existing Hillje 345-kV Substation:
 - Expand breaker-and-a-half configuration by adding two (2) new bays
 - Install five (5) Circuit Breakers in expanded breaker-and-a-half configuration
- At Hillje 765/345-kV Substation Install:
 - Three (3) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency

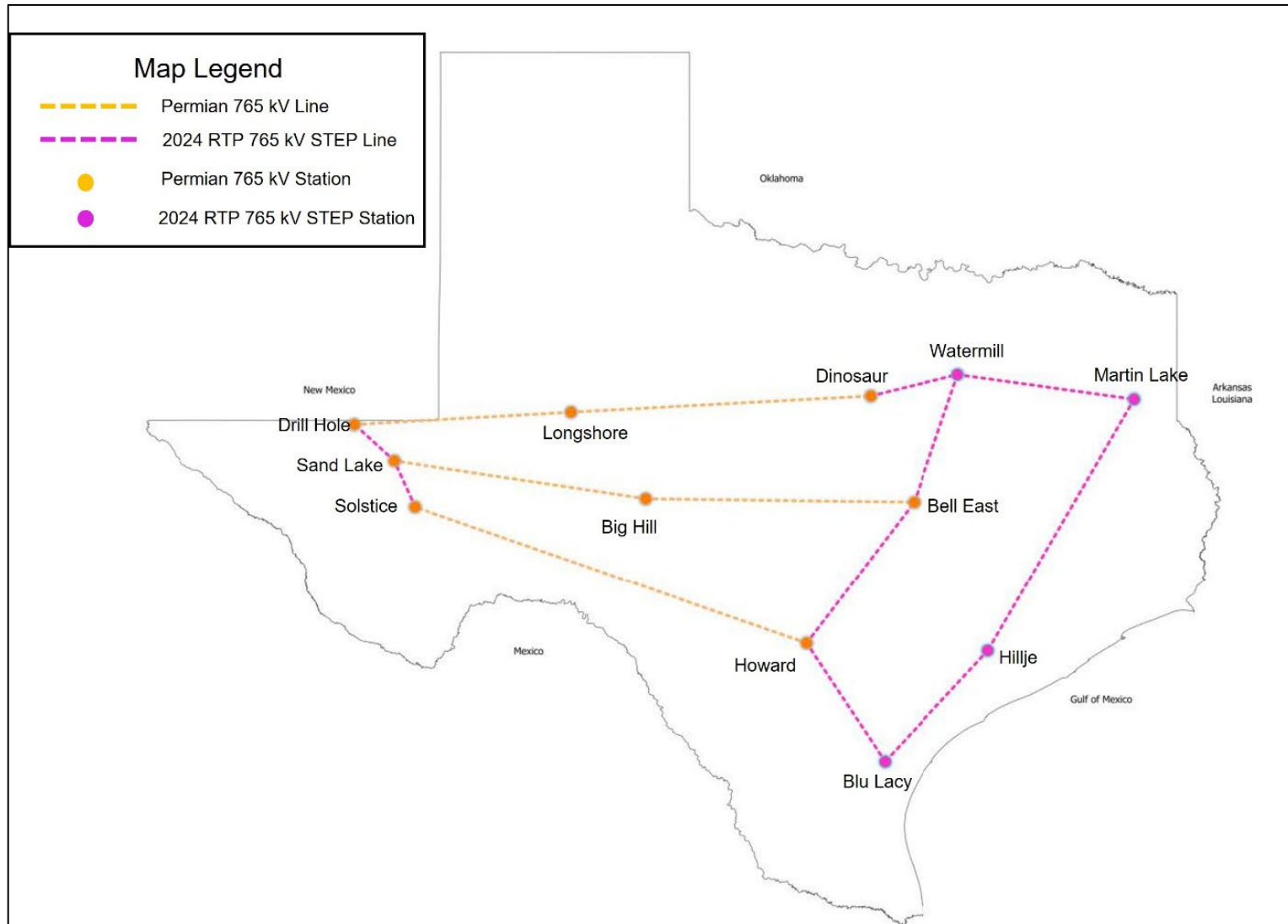
Project Proposed by AEP, CPS, Oncor and CEHE (Cont.)

- Install two (2) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at New Substation 2 [Dinosaur]
- Install reactive device on 765-kV line at New Substation 2 [Dinosaur]
- Install four (4) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at East Bell Substation
- Install reactive device on 765-kV line at East Bell Substation
- Install four (4) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at Howard Substation
- Install two (2) 300 MVAR 765-kV Line Shunt Reactors at Howard Substation
- Install one (1) 300 MVAR 765-kV Bus Shunt Reactor at Howard Substation

Project Proposed by AEP, CPS, Oncor and CEHE (Cont.)

- Construct a new Dinosaur to Watermill 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new right of way (ROW), approximately 78.0-mile;
- Construct a new Watermill to Martin Lake 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 152.4-mile;
- Construct a new Martin Lake to Hillje 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 292.8-mile;
- Construct a new Hillje to Blu Lacy 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 142.8-mile;
- Construct a new Blu Lacy to Howard 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 139.2-mile;
- Construct a new Howard to Bell East 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 170.4-mile; and
- Construct a new Bell East to Watermill 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 133.2-mile.

765-kV STEP Core Plan Proposed in 2024 RTP



Study Assumptions and Methodology

- The reliability need of this RPG project was identified and included in the 2024 RTP
- This RPG project aligns with the 2024 RTP 765-kV STEP Core Plan
- ERCOT will utilize the 2024 RTP study for this Independent Review and will assess the need for further analysis
 - Steady-state reliability analysis
 - Congestion Analysis
- ERCOT will perform additional evaluations in accordance with NERC TPL-001-5.1 and ERCOT Planning Criteria
 - Generation Addition and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
 - Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)

Next Steps and Tentative Timeline

- Tentative Timelines
 - Status update at October RPG meeting
 - Final recommendation – Q4 2025

Thank you!



Stakeholder comments also welcomed through:

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